

MICROSOFT AZURE ASSIGNMENT

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1. Microsoft Azure Directory vs. Microsoft Entra ID

Introduction

Microsoft Azure is a cloud computing platform that provides services such as virtual machines, storage, databases, networking, security, and identity management. Identity management is important because organizations need to control who can access their applications and resources.

The terms **Azure Directory** and **Microsoft Entra ID** can sometimes be confusing. Microsoft Entra ID is the current name of the service that was previously called **Azure Active Directory (Azure AD)**.

Microsoft Entra ID is Microsoft's cloud-based identity and access management service. It is used to manage identities and control access to applications, data, and resources.

Azure Directory / Azure Active Directory

"Azure Directory" is commonly used as an informal term for the directory associated with an Azure environment. The previous official name of the service was **Azure Active Directory (Azure AD)**.

An Azure directory contains information about users, groups, applications, devices, and other identity-related objects. It provides an identity boundary called a **tenant** for an organization.

For example, a college can have a Microsoft Entra tenant containing:

- Students
- Teachers
- Administrators
- Student groups
- Applications
- Registered devices

Microsoft Entra ID

Microsoft Entra ID is the current official name of Azure Active Directory. It provides identity and access management capabilities such as:

- User and group management
- Authentication
- Single Sign-On (SSO)
- Multifactor Authentication (MFA)
- Application access
- Device identity
- Conditional Access

- Role-based access control
- Managed identities

Microsoft Entra ID therefore provides a central system through which an organization can authenticate users and control their access to cloud resources.

Comparison

Azure Directory / Azure AD	Microsoft Entra ID
Older terminology	Current official name
Previously called Azure Active Directory	Current Microsoft product name
Provides identity management	Provides identity and access management
Supports users and groups	Supports users, groups and other identities
Supports authentication	Supports authentication and security policies
Supports application access	Supports application and resource access

The important point is that **Microsoft Entra ID is not a completely different directory from Azure AD**. It is the current name for the Azure Active Directory identity service. Microsoft documentation also continues to describe Azure AD as the former name of Microsoft Entra ID.

Importance of Microsoft Entra ID

Microsoft Entra ID helps organizations securely manage identities. For example, a university can use it to allow students and teachers to sign in using organizational accounts while controlling which resources they can access.

It also supports security features such as MFA and Conditional Access. This allows organizations to add additional security requirements before granting access.

Conclusion

Microsoft Entra ID is an essential part of the Microsoft Azure ecosystem. The term Azure Active Directory or Azure AD refers to its former name, while Microsoft Entra ID is the current name. It provides centralized identity and access management for users, applications, devices, and Azure resources.

2. Basics of Linux Commands

Introduction

Linux is an open-source operating system widely used for servers, cloud computing, networking, programming, and cybersecurity. Linux is also commonly used with Microsoft Azure Virtual Machines.

Azure allows users to create Linux virtual machines and manage them through tools such as SSH and Azure CLI. Microsoft also supports Microsoft Entra authentication for Linux virtual machines, allowing users to use Entra identities and Azure RBAC for VM access.

A **Linux command** is an instruction entered into the terminal to perform an operation such as managing files, navigating directories, checking system information, or installing software.

Common Linux Commands

1. pwd – Print Working Directory

Displays the current directory.

```
pwd
```

2. ls – List

Displays files and directories.

```
ls
```

Useful options:

```
ls -l
```

```
ls -a
```

-l displays detailed information, while -a also displays hidden files.

3. cd – Change Directory

Used to move between directories.

```
cd Documents
```

To move to the parent directory:

```
cd ..
```

4. mkdir – Make Directory

Creates a new directory.

```
mkdir project
```

5. touch

Creates a new empty file.

```
touch notes.txt
```

6. cat

Displays the contents of a file.

```
cat notes.txt
```

7. cp – Copy

Copies a file.

```
cp notes.txt backup.txt
```

8. mv – Move

Moves or renames a file.

```
mv notes.txt study.txt
```

9. rm – Remove

Deletes a file.

```
rm notes.txt
```

10. rmdir

Removes an empty directory.

```
rmdir project
```

11. clear

Clears the terminal screen.

```
clear
```

12. whoami

Displays the currently logged-in username.

```
whoami
```

13. date

Displays the current date and time.

```
date
```

14. sudo

Runs a command with elevated administrative privileges.

```
sudo apt update
```

15. apt

Used to manage software packages on Debian- and Ubuntu-based Linux systems.

```
sudo apt update
```

```
sudo apt install nginx
```

Linux Commands in Azure

Linux commands are useful when working with an Azure Linux Virtual Machine. After connecting to the VM using SSH, an administrator can use normal Linux commands to manage files, install applications, configure services, and monitor the system.

For example, an administrator could install a web server using:

```
sudo apt update
```

```
sudo apt install nginx
```

Microsoft Entra ID can also be integrated with Azure Linux VMs. With appropriate Azure RBAC roles, users can sign in to Linux VMs using Microsoft Entra authentication.

Conclusion

Learning basic Linux commands is important for cloud administrators and developers. Commands such as `ls`, `cd`, `mkdir`, `cp`, `mv`, `rm`, and `sudo` provide the basic skills needed to manage Linux systems and Azure Linux virtual machines.

3. Types of Identity

Introduction

An **identity** represents a person, application, device, or service that can be authenticated and authorized to access resources. Identity management is one of the important functions of Microsoft Entra ID.

Microsoft Entra ID supports different identities because cloud environments contain not only human users but also applications, devices, and automated services.

1. User Identity

A **user identity** represents an individual person.

Examples include:

- Student
- Teacher
- Employee
- Administrator
- Developer

A user can sign in to applications and access resources according to the permissions assigned to the account.

2. Guest Identity

A **guest identity** represents an external person who has been invited to collaborate with an organization.

Examples include:

- External trainer
- Consultant
- Partner
- Vendor

Guest identities are useful when an organization needs to give controlled access to people outside the organization.

3. Service Principal

A **service principal** represents an application or service that needs to access Azure resources.

For example, an automated application may need access to Azure Storage. Instead of using a human user's account, the application can use its own identity.

Service principals are useful for:

- Automation
- Scripts
- Applications
- DevOps processes

4. Managed Identity

A **managed identity** is an identity provided for an Azure resource so that applications can authenticate to other Azure services without storing credentials manually.

For example, an Azure Virtual Machine can use a managed identity to access another Azure resource.

This reduces the need to store passwords, keys, or secrets in application code. Microsoft describes managed identities as identities that applications can use to obtain access tokens for Azure resources that support Microsoft Entra authentication.

5. Device Identity

A **device identity** represents a device such as a laptop, desktop, or mobile device.

Organizations can use device identities to control access based on whether a device is known, registered, or compliant with organizational requirements.

6. Workload Identity

A **workload identity** represents an application or automated workload that requires access to cloud resources.

It is particularly useful in modern cloud applications, automation, and containerized environments.

Summary

Identity	Represents	Example
User	Individual person	Student
Guest	External person	Consultant
Service Principal	Application/service	Automation program
Managed Identity	Azure resource identity	Azure VM
Device Identity	Physical device	Laptop
Workload Identity	Application workload	Cloud application

Importance of Identity Management

Identity management allows organizations to determine **who or what is requesting access** and whether that identity should be allowed to perform the requested operation.

For example, a student may only need to read educational material, while a teacher may need additional permissions to upload or modify content.

This supports the **principle of least privilege**, where identities receive only the permissions required to perform their tasks.

Conclusion

Different identity types are necessary because cloud environments contain people, applications, devices, and automated services. Microsoft Entra ID provides a centralized platform for managing these identities and controlling their access to resources.

4. Difference Between User, Group and Role

Introduction

Users, groups, and roles are important concepts in Azure identity and access management. They are related but serve different purposes.

A user represents an individual identity, a group contains multiple identities, and a role defines the permissions that an identity can have.

User

A **user** represents one individual.

For example:

Student01

Teacher01

Admin01

Each user can have their own identity and can be assigned permissions to access resources.

Group

A **group** is a collection of users or other supported identities.

For example, a college can create a group called:

BCA-Students

and add all BCA students to it.

Instead of assigning permissions separately to every student, an administrator can assign access to the group.

Groups therefore make identity management easier, especially when an organization has many users.

Role

A **role** defines what actions an identity is allowed to perform.

Azure provides role-based access control (RBAC). Azure RBAC allows roles to be assigned to principals such as users, groups, service principals, managed identities, and workload identities.

Some common Azure roles include:

Reader

A Reader can view resources but cannot normally make changes.

Contributor

A Contributor can manage Azure resources but does not have the same permission to manage access as an Owner.

Owner

An Owner has extensive management permissions, including the ability to manage access to resources.

Difference

User	Group	Role
Represents one person	Represents a collection of users	Represents permissions
Example: Student01	Example: BCA Students	Example: Reader
Identifies an individual	Simplifies management	Controls allowed actions
Can be assigned a role	Can be assigned a role	Defines access permissions

Example

Consider a college using Azure.

There are 100 students and 10 teachers. Instead of managing every account separately, the administrator can create:

BCA-Students

BCA-Teachers

Students can be added to the first group and teachers to the second.

The administrator can then assign suitable roles:

BCA-Students → Reader

BCA-Teachers → Contributor

The basic relationship can be represented as:

User

↓

Group



Role



Azure Resource

For example:

Student01



BCA-Students



Reader



Azure Storage

This approach makes access management easier and more scalable.

Importance of Role-Based Access Control

RBAC helps organizations avoid giving users unnecessary permissions. A user should receive only the access required for their work.

For example, a student who only needs to view a resource does not need administrator permissions. Similarly, a developer may need permission to manage a resource but not necessarily permission to manage user access.

This improves security and reduces the possibility of accidental or unauthorized changes.

Conclusion

Users, groups, and roles have different functions. A **user** identifies an individual, a **group** organizes multiple identities, and a **role** defines the permissions available to an identity.

Using them together provides a structured and secure way to manage access to Azure resources.

FINAL CONCLUSION

Microsoft Azure provides powerful cloud services, and identity and access management are essential for protecting these services. Microsoft Entra ID provides centralized identity management and controls access to applications, data, and resources.

Linux is also important in Azure because Linux virtual machines are widely used for cloud applications and server environments. Basic Linux commands allow administrators to manage files, directories, software, and system operations.

Different identity types allow Azure to manage people, applications, devices, and automated workloads. Finally, users, groups, and roles work together to provide controlled access to Azure resources through role-based access control.

Therefore, understanding **Microsoft Entra ID, Linux commands, identity types, users, groups, and roles** provides an important foundation for working with Microsoft Azure and cloud security.

References

1. Microsoft Learn – Microsoft Entra ID documentation.
2. Microsoft Learn – Azure Role-Based Access Control (RBAC).
3. Microsoft Learn – Sign in to Linux Virtual Machines using Microsoft Entra ID.
4. Microsoft Learn – Managed Identities for Azure Resources.